Continuous Active Inference for 3-D Trajectory Tracking

Algorithm Description & Mathematical Framework

Contents

1 Overview **2**

2 State & Action Spaces **2**
2.1 State vector 2
2.2 Action vector 2

3 Generative Process (True World) **2**

4 Generative Model (Agent's Internal Model) **3**
4.1 Prior / Reference trajectory 3

5 Free Energy & Precision Matrices **3**
5.1 Variational free energy 3
5.2 Precision matrices 3

6 Belief Update (Perception) **4**

7 Action Policy (Active Inference) **4**

8 Algorithm **5**

9 System Diagram **5**

10 Parameter Summary **6**

11 Key Properties & Interpretation **6**

A Discrete Integration **7**

1 Overview

This document describes a **continuous-time Active Inference** (AIF) agent that tracks a 3-D trajectory under gravity. The system illustrates a key principle of the Free Energy Principle: axes that are *neither observed nor modelled* accumulate unbounded uncertainty and drift — the agent cannot and does not attempt to correct them.

Core design asymmetry.

- x**-axis** (horizontal): controlled + visual feedback.
- y**-axis** (vertical): controlled + proprioceptive / visual feedback; gravity compensation built in.
- z**-axis** (depth): *uncontrolled, no feedback* — undergoes a random walk driven by unmodelled noise.

2 State & Action Spaces

2.1 State vector

The full state is

$$\boldsymbol{x} = \begin{bmatrix} p_1, & p_2, & p_3, & v_1, & v_2, & v_3 \end{bmatrix}^\top \in \mathbb{R}^6, \tag{1}$$

where (p_1, p_2, p_3) are positions along (x, y, z) and (v_1, v_2, v_3) the corresponding velocities.

2.2 Action vector

$$\boldsymbol{a} = [u_1,\ u_2]^\top \in \mathbb{R}^2, \qquad u_3 \equiv 0. \tag{2}$$

No action is generated along z because the free-energy gradient $\partial\mathcal{F}/\partial u_3 = 0$ (Sec. 5).

3 Generative Process (True World)

The continuous-time dynamics of the real world are

$$\dot{p}_1 = v_1, \qquad \dot{p}_2 = v_2, \qquad \dot{p}_3 = v_3, \tag{3}$$

$$\dot{v}_1 = \frac{u_1}{m_1} + \omega_x(t), \qquad \dot{v}_2 = \frac{u_2}{m_2} - g + \omega_y(t), \qquad \dot{v}_3 = \omega_z(t), \tag{4}$$

where $g = 9.81\ \mathrm{m/s^2}$ and ω_i are independent Gaussian white-noise processes,

$$\omega_i(t) \sim \mathcal{N}\left(0,\ \frac{\sigma_i^2}{\Delta t}\right), \quad i \in \{x, y, z\}. \tag{5}$$

Why z drifts. Because $\dot{v}_3 = \omega_z(t)$ and $u_3 = 0$, the z-velocity performs a Wiener process (Brownian motion) with diffusion coefficient σ_z:

$$v_3(t) = v_3(0) + \int_0^t \omega_z(s)\,\mathrm{d}s, \qquad \mathrm{Var}[v_3(t)] = \sigma_z^2 t.$$

Integrating once more, p_3 follows a second-order random walk. This mimics the inertia-induced depth drift observed in arm-movement experiments.

Observation model. Only the x-y components are sensed:

$$\boldsymbol{y}_{\text{obs}} = \big[p_1, p_2, \underbrace{-}_{?}, v_1, v_2, \underbrace{-}_{?}\big]^\top + \boldsymbol{\eta}, \quad \boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma}_{\text{obs}}), \tag{6}$$

where $-$ denotes unobserved entries; in practice those slots are set equal to the current belief $\boldsymbol{\mu}$ so that the sensory prediction error on z is identically zero.

4 Generative Model (Agent's Internal Model)

The agent assumes z is *stationary*:

$$f_{\text{model},\,p_3}(\boldsymbol{x}, \boldsymbol{a}) = 0, \tag{7}$$

$$f_{\text{model},\,v_3}(\boldsymbol{x}, \boldsymbol{a}) = 0. \tag{8}$$

For the controlled axes the model mirrors the true physics:

$$\boldsymbol{f}_{\text{model}}(\boldsymbol{x}, \boldsymbol{a}) = \big[v_1,\ v_2,\ 0,\ u_1/m_1,\ u_2/m_2 - g,\ 0\big]^\top. \tag{9}$$

Combined with $\Pi_{s,z} = 0$ (no z-observations), the belief about z is completely frozen at its initial value — the agent *correctly* does nothing about a dimension it cannot observe or model.

4.1 Prior / Reference trajectory

$$\boldsymbol{x}^*(t) = \big[p_1^*(t),\ p_2^*,\ 0,\ v_1^*,\ 0,\ 0\big]^\top, \tag{10}$$

with a linear ramp in x:

$$p_1^*(t) = x_{\text{start}} + (x_{\text{end}} - x_{\text{start}})\ \min\Big(\tfrac{t}{T_{\text{end}}}, 1\Big). \tag{11}$$

5 Free Energy & Precision Matrices

5.1 Variational free energy

$$\mathcal{F}(\boldsymbol{\mu}, \boldsymbol{a}) = \underbrace{\tfrac{1}{2}\,\boldsymbol{\varepsilon}_p^\top\,\boldsymbol{\Pi}_p\,\boldsymbol{\varepsilon}_p}_{\text{prior cost}} + \underbrace{\tfrac{1}{2}\,\boldsymbol{\varepsilon}_s^\top\,\boldsymbol{\Pi}_s\,\boldsymbol{\varepsilon}_s}_{\text{sensory cost}}, \tag{12}$$

where the prediction errors are

$$\boldsymbol{\varepsilon}_p = \boldsymbol{\mu} - \boldsymbol{x}^*(t), \qquad \boldsymbol{\varepsilon}_s = \boldsymbol{y}_{\text{obs}} - \boldsymbol{\mu}. \tag{13}$$

5.2 Precision matrices

Both $\boldsymbol{\Pi}_p$ and $\boldsymbol{\Pi}_s$ are 6×6 diagonal matrices with structure:

$$\boldsymbol{\Pi}_s = \operatorname{diag}\Big(\tfrac{1}{\sigma_{s,\text{pos}}^2},\ \tfrac{1}{\sigma_{s,\text{pos}}^2},\ \mathbf{0},\ \tfrac{1}{\sigma_{s,\text{vel}}^2},\ \tfrac{1}{\sigma_{s,\text{vel}}^2},\ \mathbf{0}\Big), \tag{14}$$

$$\boldsymbol{\Pi}_p = \operatorname{diag}\Big(\tfrac{1}{\sigma_{p,\text{pos}}^2},\ \tfrac{1}{\sigma_{p,\text{pos}}^2},\ \tfrac{1}{\sigma_{p,z}^2} \approx 0,\ \tfrac{1}{\sigma_{p,\text{vel}}^2},\ \tfrac{1}{\sigma_{p,\text{vel}}^2},\ \tfrac{1}{\sigma_{p,z}^2} \approx 0\Big). \tag{15}$$

Why no z-action is generated. The action gradient $\partial\mathcal{F}/\partial u_3$ involves $\boldsymbol{\Pi}_s\,\boldsymbol{\varepsilon}_s$ and $\boldsymbol{\Pi}_p\,\boldsymbol{\varepsilon}_p$ projected onto the z-rows. Since $\Pi_{s,z} = 0$ (zero sensory precision on z) and $\Pi_{p,z} \approx 0$

(negligible prior precision on z), this gradient is identically zero:

$$\frac{\partial \mathcal{F}}{\partial u_3} = 0 \quad \Rightarrow \quad u_3 \equiv 0.$$

6 Belief Update (Perception)

The agent's belief $\boldsymbol{\mu}$ is updated by gradient descent on free energy, corrected by the model's own flow:

$$\dot{\boldsymbol{\mu}} = \boldsymbol{f}_{\text{model}}(\boldsymbol{\mu}, \boldsymbol{a}) + \kappa_\mu \big(\boldsymbol{\Pi}_s \, \boldsymbol{\varepsilon}_s - \boldsymbol{\Pi}_p \, \boldsymbol{\varepsilon}_p\big), \tag{16}$$

with learning rate $\kappa_\mu > 0$. The first term propagates the belief forward according to the agent's dynamics; the bracketed term drives the belief towards the observation (sensory term) and away from the prior error (prior term).

z-axis behaviour. For the z-components:

- $f_{\text{model},z} = 0$ (stationary model assumption),
- $\Pi_{s,z} \, \varepsilon_{s,z} = 0$ (no z-observation),
- $\Pi_{p,z} \, \varepsilon_{p,z} \approx 0$ (weak z-prior).

Hence $\dot{\mu}_z \approx 0$: the z-belief is frozen.

7 Action Policy (Active Inference)

The control law minimises $\mathcal{F}$ with respect to $\boldsymbol{a}$. Because the sensory precision on z is zero, this reduces to a PD controller on the x-y plane only:

$$u_1 = m_1 \big[K_P \, (p_1^* - p_1) + K_D \, (v_1^* - v_1)\big], \tag{17}$$

$$u_2 = m_2 \big[K_P \, (p_2^* - p_2) + K_D \, (v_2^* - v_2)\big] + m_2 g, \tag{18}$$

clipped to $[-a_{\max}, a_{\max}]$. The gravity feed-forward term $m_2 g$ in u_2 emerges naturally from including g in the generative model.

8 Algorithm

Algorithm 1 Continuous Active Inference — 3-D Trajectory Tracking

Require: Parameters $\{m_1, m_2, g, K_P, K_D, \kappa_\mu, \Delta t, T_{\text{end}}\}$, precision matrices $\boldsymbol{\Pi}_p$, $\boldsymbol{\Pi}_s$, noise levels $\{\sigma_x, \sigma_y, \sigma_z\}$

Initialise: $\boldsymbol{x}_{\text{real}} \leftarrow \boldsymbol{x}_0$; $\quad \boldsymbol{\mu} \leftarrow \boldsymbol{x}_0$
Pre-draw noise arrays: $\omega_i[k] \sim \mathcal{N}(0,\, \sigma_i^2/\Delta t)$ for $i \in \{x, y, z\}$, $k = 0, \ldots, N-1$
for $k = 0, 1, \ldots, N-1$ **do**
  $t \leftarrow k\,\Delta t$
  Reference: $\boldsymbol{x}^*(t) \leftarrow \textsc{DesiredState}(t)$

  — **Perception** —
  Observe $\boldsymbol{y}_{\text{obs}}$: set $y_{p_3} \leftarrow \mu_{p_3}$, $y_{v_3} \leftarrow \mu_{v_3}$ $\quad (\Rightarrow \varepsilon_{s,z} = 0)$
  Compute prediction errors: $\boldsymbol{\varepsilon}_p \leftarrow \boldsymbol{\mu} - \boldsymbol{x}^*$; $\quad \boldsymbol{\varepsilon}_s \leftarrow \boldsymbol{y}_{\text{obs}} - \boldsymbol{\mu}$
  Compute free energy: $\mathcal{F} \leftarrow \frac{1}{2}\boldsymbol{\varepsilon}_p^\top \boldsymbol{\Pi}_p \boldsymbol{\varepsilon}_p + \frac{1}{2}\boldsymbol{\varepsilon}_s^\top \boldsymbol{\Pi}_s \boldsymbol{\varepsilon}_s$

  — **Action** —
  $u_1 \leftarrow m_1[K_P(p_1^* - \mu_{p_1}) + K_D(v_1^* - \mu_{v_1})]$
  $u_2 \leftarrow m_2[K_P(p_2^* - \mu_{p_2}) + K_D(v_2^* - \mu_{v_2})] + m_2 g$
  Clip $u_1, u_2 \in [-a_{\max}, a_{\max}]$; $\quad u_3 \leftarrow 0$

  — **World update** —
  $\dot{\boldsymbol{x}}_{\text{real}} \leftarrow \boldsymbol{f}_{\text{proc}}(\boldsymbol{x}_{\text{real}}, \boldsymbol{a}, [\omega_x[k], \omega_y[k], \omega_z[k]])$
  $\boldsymbol{x}_{\text{real}} \leftarrow \boldsymbol{x}_{\text{real}} + \dot{\boldsymbol{x}}_{\text{real}}\,\Delta t$

  — **Belief update** —
  $\dot{\boldsymbol{\mu}} \leftarrow \boldsymbol{f}_{\text{model}}(\boldsymbol{\mu}, \boldsymbol{a}) + \kappa_\mu\big(\boldsymbol{\Pi}_s \boldsymbol{\varepsilon}_s - \boldsymbol{\Pi}_p \boldsymbol{\varepsilon}_p\big)$
  $\boldsymbol{\mu} \leftarrow \boldsymbol{\mu} + \dot{\boldsymbol{\mu}}\,\Delta t$

  Store $\boldsymbol{x}_{\text{real}}$, $\boldsymbol{\mu}$, $\boldsymbol{x}^*$, $\boldsymbol{a}$, $\mathcal{F}$
end for

9 System Diagram

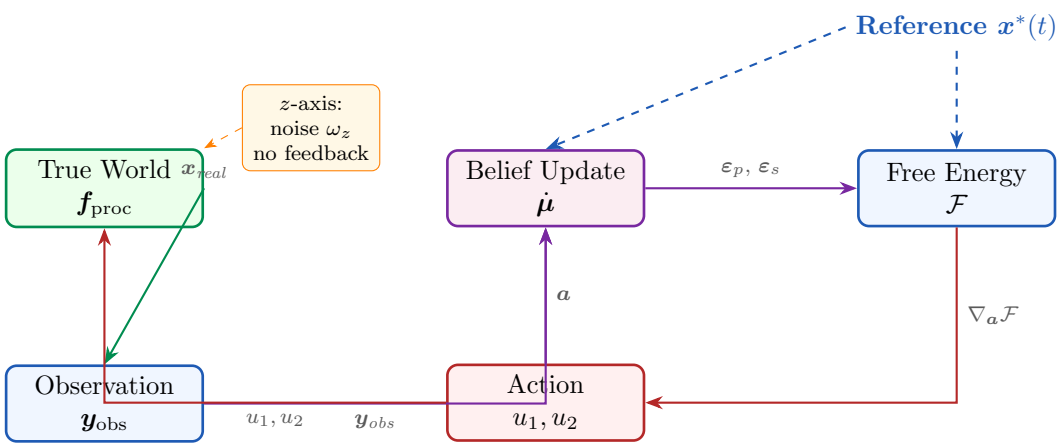

Figure 1: Active Inference loop. Green: world dynamics; purple: perception/belief; red: action policy; orange: uncontrolled z-perturbation (no feedback path exists).

10 Parameter Summary

Table 1: Default simulation parameters.

Symbol	Description	Default value
m_1, m_2	Effective masses (x, y)	1.0 kg
g	Gravitational acceleration	9.81 m/s^2
K_P	Proportional gain	20.0
K_D	Derivative gain	12.0
$a_{\max}$	Action clip bound	50.0 N
κ_μ	Belief learning rate	1.0
Δt	Integration step	0.01 s
T_{end}	Episode duration	4.0 s
$\sigma_{s,\text{pos}}$	Sensory pos. precision (x,y)	0.1
$\sigma_{s,\text{vel}}$	Sensory vel. precision (x,y)	0.15
$\sigma_{s,z}$	Sensory z precision	0 (unobserved)
$\sigma_{p,\text{pos}}$	Prior pos. precision (x,y)	0.1
$\sigma_{p,z}$	Prior z precision	$1.0 \gg \sigma_{p,\text{pos}}$
σ_z	Process noise on z-velocity	tunable

11 Key Properties & Interpretation

1. **Asymmetric control.** The x-y plane is fully controlled via PD action derived from free-energy minimisation. The z-axis is intentionally ignored at every level: no model, no observation, no action.

2. **Gravity compensation.** Because gravity appears in both the generative process and the generative model, the feed-forward term $m_2 g$ in u_2 cancels it exactly, keeping $p_2 \approx p_2^*$ at all times.

3. **Frozen z-belief as a correct AIF solution.** Setting $\Pi_{s,z} = 0$ is not a hack — it is the principled Bayesian answer to "I have no information about z". The belief update equation then gives $\dot{\mu}_z = 0$, so the agent neither tracks nor fights the z-drift.

4. **Free energy as a dual objective.** $\mathcal{F}$ simultaneously encodes trajectory tracking (via $\mathbf{\Pi}_p$, which punishes deviations from $\boldsymbol{x}^*$) and sensory consistency (via $\mathbf{\Pi}_s$, which punishes deviations between observations and belief). Increasing $\mathbf{\Pi}_p$ relative to $\mathbf{\Pi}_s$ tightens trajectory adherence; increasing $\mathbf{\Pi}_s$ makes the agent more reactive to sensory input.

5. **Extensibility.** The framework naturally accommodates:

   - Non-linear or time-varying reference trajectories (replace `desired_full_state`).
   - Salient stimuli: add a salience-weighted term to $\mathcal{F}$ that attracts the reference or modulates $\mathbf{\Pi}_s$ dynamically.
   - Partial z-feedback: simply set $\Pi_{s,z} > 0$ and extend $\boldsymbol{y}_{\text{obs}}$ to include p_3; the belief will begin tracking and correcting z-drift.

A Discrete Integration

All ODEs are integrated with the forward Euler scheme:

$$\boldsymbol{x}(t + \Delta t) = \boldsymbol{x}(t) + \boldsymbol{f}\big(\boldsymbol{x}(t), \boldsymbol{a}(t)\big)\,\Delta t. \tag{19}$$

For the stochastic noise, each sample is scaled as $\omega_i \sim \mathcal{N}(0,\, \sigma_i^2/\Delta t)$ so that the *integrated* increment over one step has variance $\sigma_i^2\,\Delta t$, consistent with a Wiener process of diffusion rate σ_i.